

12.1 INTRODUCTION

The word subsidence usually refers to the downward movement of a ground surface, or "settlement" or "depression" due to caving of underground rock strata. Subsidence is the "result" of activities done in underground. These activities can be manmade or natural. The most common causes of subsidence are: (i) withdrawal of fluids from pores and fractures, and (ii) disturbance of the ground by excavation of minerals. In the first case, pumping of ground water from aquifers and consequent drawdown of the water table is a very common cause of surface subsidence or depression. Withdrawal of oil, and to a lesser extent, withdrawal of gas from unconventional reservoir is also a significant source of subsidence in some areas. In these cases, the overburden strata drawdown due to a consolidation process after withdrawal of fluids. Secondly, surface and underground excavation cause subsidence through induced changes in stress, strain, and displacement fields in adjacent soil and rock masses. Sometimes excavation may be a natural process, for example, formation of solution cavities by circulating ground water or flowing water channel in limestone formations. If the induced displacements from underground excavation extend to the surface, then subsidence occurs on the surface. In cases where damage to surface structures or underground utilities occur, there is considerable importance attached to the determination of the cause of damage in the presence of subsidence and the actual subsidence mechanism.

Caving is an often observed phenomenon associated with underground mining and begins with collapse of the back, roof, or hanging wall. Caving may occur unexpectedly or be deliberately induced based on the rock strata and mining methods such as block caving, sub-level caving, longwall mining, bord and pillar mining with pillar extraction. Caving begins with rock fractures and failures just above the excavated zone and extends to the upper rock layers. Finally, broken rock pieces fall to the floor. Failure of the rock mass is greatly assisted by the presence of pre-existing joints, slip planes, shear zones and other discontinuities. Caving process may cease once the caved bulk volume completely occupies the void space and an equilibrium condition occurs. It may be noted that volume of broken rock mass is higher than

the *in situ* rock mass. This ratio of broken rock mass to that of *in situ* rock mass is termed as *bulking factor*.

Subsidence profile and its formation depend on type of method of mining, proximity of underground excavation from the surface and condition of overlaying rock strata. In general, following types of subsidence have been observed:

(i) Trough subsidence
(ii) Subsurface caving
(iii) Plug caving
(iv) Chimneying or piping or funnelling or sinkhole

Subsidence over a mine field can be defined as the vertical movement of the ground surface and is the result of re-adjustments of the overburden due to collapse or failure of underground mine workings. Surface subsidence features usually take the form of either sinkholes or troughs. In this chapter, the mechanics of trough subsidence and sinkhole are discussed.

12.2 CONCEPT OF CAVING AND FORMATION OF SUBSIDENCE

Trough type of subsidence starts with the caving of underground rock strata. The *caving zone* is defined as the zone where broken rock materials are fallen and compacted as shown in Figure 12.1. The process of caving has been described in the previous section. Rock layers just above the caving zone settle on top of the broken material having large displacements and strains. They are separated from each other and hence, multiple fractures develop in rock strata. This process continues for a few rock strata above the caving zone and it termed as *fractured zone* (Figure 12.1). Above the fractured zone, rock layers may sag having a continuous deformation with minimal cracks or fractures. They behave like beams bended under own load and load from the overlaying strata. This zone is termed as *continuous deformation zone*. On top of this zone, soil layer may deform causing trough subsidence on the surface.

Caving process for chimney or sinkhole subsidence is quite different and will be discussed in Section 12.5.

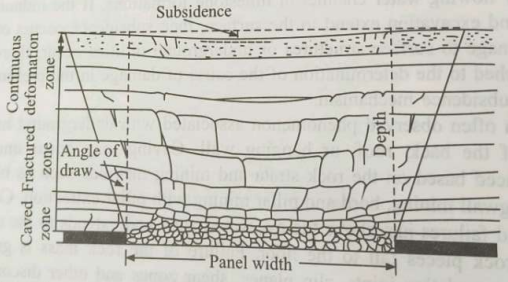


FIGURE 12.1 Schematic of caving subsidence over a longwall panel showing fracture zones (adapted from Peng, 1992).

12.3 TROUGH SUBSIDENCE

The movement of the ground surface results into both horizontal and vertical displacements. The vertical displacement is generally the maximum at the middle of the excavation. The magnitude of subsidence gradually becomes zero outside the limit of the excavation. Figure 12.2 shows a schematic drawing of influenced zone generated on the surface due to formation of trough over a mined out longwall panel. This influenced zone is termed as *subsidence basin*. The figure also shows the major cross-section and longitudinal section taken along the middle of the subsidence basin. The trough created is confined between the limit lines shown as angle drawn with subsidence occurring outside the excavated area. Traditionally the subsidence profile is drawn with the vertical component of ground movement. Horizontal displacements act towards the area of maximum subsidence and the magnitude are approximately a function of gradient of the subsidence profile. At the centre of the mined out area, the horizontal displacement is zero.

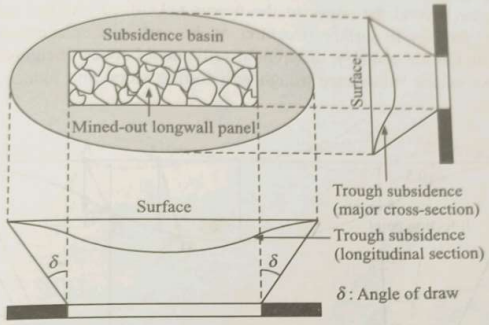


FIGURE 12.2 Subsidence basin and cross-sections.

The magnitude, extent and shape of a trough subsidence basin and surface deformation ry from mine to mine and from panel to panel. Overlaying strata above a longwall panel e normally divided into two categories:

(i) *Immediate roofs:* upto 5 to 10 m above the seam
(ii) *Main roofs:* upto 20 to 30 m above the immediate roofs.

e factors of immediate and main roofs and others, which affect these properties, are:

(i) Physical and geomechanical properties of immediate and main roofs strata

Immediate roofs	*Main roofs*
• Strong/hard	Strong/hard
• Strong/hard	Weak/soft
• Weak/soft	Strong/hard
• Weak/soft	Weak/soft

(ii) Mined out opening or gob size
(iii) Mining depth
(iv) Mining height
(v) Topography
(vi) Time

2.3.1 Components of Surface Movement

igure 12.3 shows typical subsidence profiles for sub-critical and super-critical subsidence over longwall panel. Based on this, the following components are defined.

(i) **Subsidence(s):** On any major cross-section, the vertical component of the surface movement vector (vector **AB** shown in Figure 12.3) is called *surface subsidence*. It generally points downward. But sometimes it points upward in areas ahead of the face line or beyond the edges of the opening. Uniform vertical subsidence does not cause any damage to surface structures as the vertical displacement at every point of a structure. On the contrary, differential settlement between neighbouring points along the surface of the subsidence trough will form a slope and induce tilting.

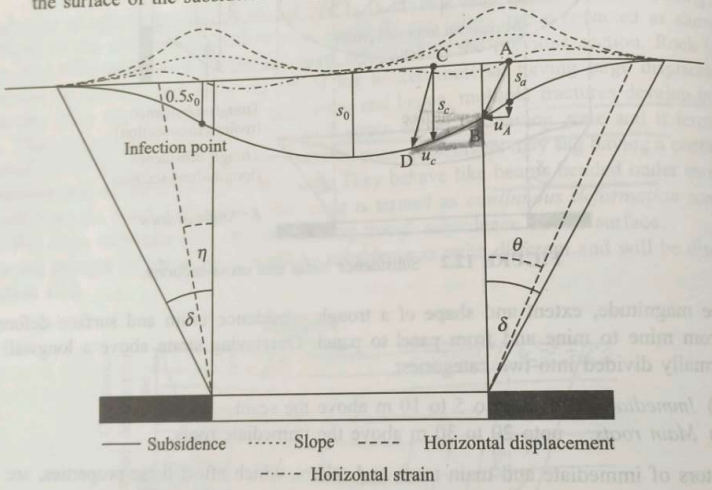


FIGURE 12.3 Terminologies used associated with subcritical subsidence profile.

(ii) **Horizontal displacement (u):** On any major cross-section, the horizontal component of the surface movement vector is called *surface horizontal displacement*. It generally points toward the centre of the subsidence basin. But in steep terrain, it moves along the down dip direction.

(iii) **Slope ($\psi = ds/dx$):** On any major cross-section, the gradient of subsidence curve at any point is denoted as slope at that point. In discrete sense, slope is estimated as the difference in surface subsidence between the two end points of a line segment divided by the horizontal distance between the two points. For example, in Figure 12.3, slope between point A and C will be

$$\psi = \frac{s_C - s_A}{x_C - x_A} \tag{12.1}$$

(iv) **Curvature ($\kappa = d^2s/dx^2$):** On any major cross-section, the gradient of slope at any point on the surface is termed as the *curvature* at that point. In discrete sense, curvature is measured as the difference in surface slope between two adjacent line segments divided by the average length of the two line segments. There are two types of curvature: (a) convex or positive curvature and (b) concave or negative curvature.

(v) **Horizontal strain ($\varepsilon = du/dx$):** On any major cross-section, the gradient of horizontal displacement at any point on the surface is termed as *horizontal strain* at that point. The difference in horizontal displacement between two end points of a line segment on the surface divided by the length of the segment is estimated to be the horizontal strain. Horizontal strain can be positive or tensile or negative or compressive. Horizontal strain between points A and C can be expressed as:

$$\varepsilon = \frac{u_C - u_A}{x_C - x_A} \tag{12.2}$$

It is noteworthy that uniform horizontal displacement may not cause any direct damage to surface structures. However, differential displacement induces compression or tensile strains along the surface of the subsidence trough. Most surface structural damage in a subsidence trough is caused by horizontal strains. When there is differential settlement, the gradient of the ground surface changes and may disrupt and/or alter drainage systems, the flow of cracks, the gravity flow of liquids in pipelines, and the grade of highways or railroad beds. Differential settlement will also induce tilting of tall buildings, towers, and chimneys and cause instability.

12.3.2 Terminologies of Subsidence Profile

(i) Point of inflection: The point that divides the concave and convex portions of the subsidence profile curve is termed as *infection point*. At the infection point, subsidence is half of the maximum subsidence value at the centre, the surface slope is maximum, the curvature is zero and the horizontal strain is zero. The location of the inflection point depends on the size of opening or goaf or rock types. When the opening is very small as compared to depth, it is located beyond the edges of the opening on the solid coal side. As the opening grows in size, the inflection point moves inside the edges of the opening. For supercritical condition, the inflection will be inside the edges of the opening as shown in Figure 12.4. The distance from the inflection point to the nearest edge of the opening is the termed as *offset* of the inflection point.

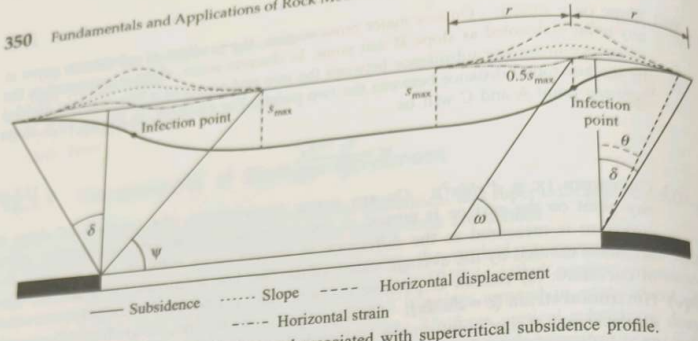


FIGURE 12.4 Terminologies used associated with supercritical subsidence profile.

(ii) Angle of draw (δ): Assuming a rectangular mined out panel, subsidence affected surface strata take the form of obtuse pyramid as mentioned in Figure 12.3. The angle subtended between the vertical line drawn at panel edge and a line joining the extraction edge or zero subsidence is termed as angle of draw or angle of limit of subsidence (Figure 12.3). Angle of draw generally ranges from 8° to 45° depending on the properties of overburden rock strata and their thickness. It is found that angle of draw is lower for stronger overburden rocks and higher for weaker rocks and soils. Pariseau (2007) mentioned that the presence of much sandstone is a likely cause of reduction in angle of draw. Sheorey et al. (2000) mentioned that the angle of draw for Indian coalfields varies from 9° to 25°. There are indication that angle of draw depend on:

- Depth
- Seam thickness
- Local geology especially major faults or fracture planes or the location of self-supporting strata above the coal seam.

(iii) Angle of critical deformation (θ): The angle between the vertical line at the opening edge and the line connecting the opening edge and the point of critical deformation on the surface is the *angle of critical deformation*. The critical deformation is defined based on the structural resistance to subsidence or horizontal strain. For example, if a structure on the surface is designed to withstand a certain amount of deformation or strain, it is obvious that angle of critical deformation of the structure will be less than angle of draw. The angle of critical deformation is on the average 5° to 10° less than the corresponding angle of draw.

(iv) Angle of break (η): On any major cross-section, the angle between the vertical line at the opening edge and the line connecting the opening edge and the point of maximum tensile strain on the surface is called the *angle of break*. The ground surface at the point of maximum tensile strain is the most likely place where tensile cracks occur.

(v) Radius (r) and angle of major influence (ω): When size of the opening or goaf reaches the critical size, the major surface deformations occur on both sides of the inflection point within a certain distance. This distance is called the *radius of major influence*. Beyond this distance, surface deformations are very small. For supercritical subsidence, infection point lies inside the panel. Then the radius of major influence is measured as the distance from the inflection to the zero subsidence (Figure 12.4). The *angle of major influence* is the angle between the horizontal and the line connecting the inflection point and the edge of the radius of major influence, mathematically, $\tan \omega = H/r$, where H is the mining depth.

(vi) Angle of full subsidence (ψ): On a major cross-section of the subsidence basin under super critical width of mining the acute angle between the horizontal and the line connecting the edge of the flat bottom of the subsidence basin and the edge of the opening is called the *angle of full subsidence*. It indicates the degree of subsidence development and can be used to define the area within which subsidence has been fully developed.

12.3.3 Estimation of Maximum Subsidence

The maximum possible subsidence (s_{max}) at the middle of the trough is directly related to the height of the excavation or mining height (m), properties of overlaying rock strata and level of stowing or backfilling. For a longwall panel having a seam dip of α, the maximum possible subsidence can be approximated as:

$$s_{max} = am \cos \alpha \tag{12.3}$$

where, a is the subsidence factor and m is the mining height. If the seam is nearly flat, the angle α is zero. It may be noted that the factor a mostly depends on the properties of overlaying strata and width to depth ratio of the panel. Subsidence factor is dimensionless and is dependent upon the (i) nature of the overburden and (ii) type of goaf treatment such as caving and some form of stowing. In general, in France, Germany and in UK, it is 0.9 while in case of stowing pneumatically (0.4–0.5) and hydraulic sand stowing (0.12). It decreases with increasing depth.

The maximum subsidence and profile thereof also depend on the width to depth ratio of a longwall panel. As shown in Figure 12.5, the width and depth have substantial effect on the surface subsidence. The width to depth ratio for which there is no subsidence, the ratio is called non-effective width (NEW). The non-effective width to depth ratio is found to vary approximately between 0.2 and 0.8 in Indian coalfields depending on strata competence. The figure shows that for particular W/H ratio the maximum subsidence is the same, although the area of influence will be more for greater depth for same angle of draw. Mathematically, it can be expressed as follows:

$$W_i' = W_i + 2H_i \tan \delta \tag{12.4}$$

In general, three types of surface subsidence can be expected based on length and width of longwall panel, overburden strata and depth of the seam. They are explained as follows:

(i) Critical area: An area of working just sufficient to cause the maximum possible subsidence (s_{max}) at one point on the surface. To get a maximum subsidence at the centre of the panel, both length and width of the opening are to be a size equal to the critical size. If any of them is

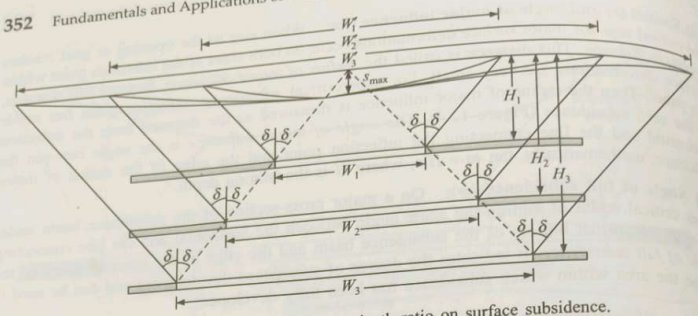


FIGURE 12.5 Effect of width to depth ratio on surface subsidence.

less, then s_{max} cannot be achieved. Let us assume, β denotes the abutment angle as mentioned in Chapter 8. Hence, critical width (W_c) [Figure 12.6(a)] equals to

$$W_c = 2H \tan \beta \tag{12.5}$$

From the data obtained from UK longwall panels, it is found that angle of draw or in this case, abutment angle is about 35°. This leads to $W_c = 1.4\ H$. The critical width for $\beta = 30°$, most common over longwall panels in former USSR and India, the critical width is $W_c = 1.15\ H$.

(ii) Subcritical area: An area of working not sufficient to cause the maximum possible subsidence at a point on the surface. The maximum subsidence s_o occurred in a subcritical subsidence basin is less than s_{max}. In this case, width to height ratio is such that $W < W_c$ [Figure 12.6(b)].

(iii) Supercritical area: An area of working causing complete subsidence at a part of the surface. In other words, s_{max} occurs at more than one point on the surface. Figure 12.6(c) clearly shows that in this case, width to depth ratio will be high and hence, $W > W_c$.

Estimation of Factor (a)

Subsidence factor is strongly depended on the properties of overburden strata, depth of mining, panel dimension and time duration. This factor can be estimated for caving longwall panel as (Peng, 1992):

$$a = 0.35 + 0.5P \tag{12.6}$$

where P is the combined strata properties and are calculated as:

$$P = \sqrt{\frac{\sum_{i=1}^{N} (Q_i h_i)^2}{\sum_{i=1}^{N} (h_i)^2}} \tag{12.7}$$

where h_i is the thickness of ith strata and Q_i corresponds to the coefficient of stratum properties as given in Table 12.1. This table also shows the rock type and uniaxial compressive strength.

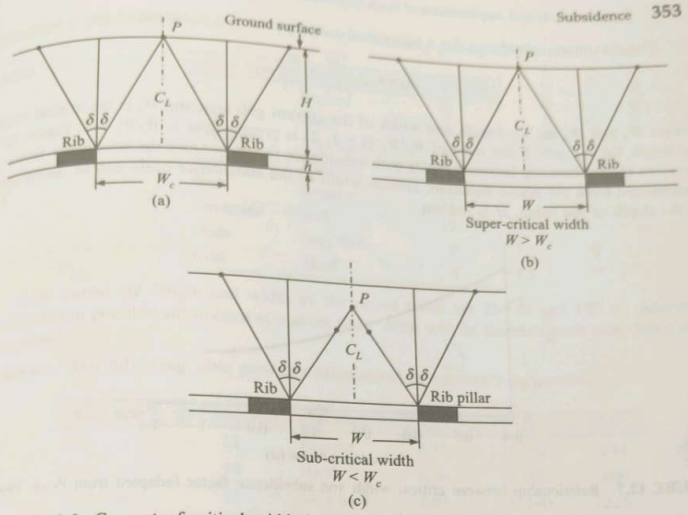


FIGURE 12.6 Concept of critical width in relation to rib-side subsidence profiles (adapted from Pariseau, 2007).

TABLE 12.1 Rock Factor Q for Different Rock Types (Peng, 1992)

Hardness		*Sandstone*	*Limestone*	*Shale*	*Coal*	*Clay*
Extremely hard	σ_c	>88	>86	>69	>27	>18
	Q	0.00	0.01	0.20	0.80	0.85
Very hard	σ_c	78 – 88	76 – 86	58 – 69	24 – 27	15 – 18
	Q	0.02	0.02	0.30	0.83	0.87
Hard	σ_c	69 – 78	68 – 86	44 – 58	20 – 24	13 – 16
	Q	0.05	0.06	0.40	0.87	0.89
Regular	σ_c	58 – 69	57 – 68	29 – 44	17 – 20	12 – 14
	Q	0.10	0.12	0.60	0.90	0.92
Soft	σ_c	48 – 58	48 – 57	19 – 29	10 – 17	7 – 12
	Q	0.2	0.22	0.7	0.93	0.95
Very soft	σ_c	44 – 48	39 – 48	10 – 19	7 – 10	3 – 7
	Q	0.30	0.40	0.80	0.97	1.00
Extremely soft	σ_c	<44	<39	<10	<7	<3 (400)
	Q	0.40	0.80	0.90	1.00	1.00

σ_c: Uniaxial compressive strength MPa; Q: Assigned rock factor.

The maximum subsidence for a subcritical subsidence basin is estimated as:

$$s_o = s_{max}\sqrt{\frac{W_1}{W_c}\times\frac{W_2}{W_c}} \tag{12.8}$$

where W_1 and W_2 are the length and width of the current gob area and W_c is the critical width or length for the gob. If the ratio of W_i/W_c (i = 1, 2) is greater than 1, $W_i/W_c = 1$. Figure 12.7 presents the relationship between W_c/H with subsidence factor, a. Once the subsidence factor is determined from the above equation, critical width of the subsidence basin can be ascertained if the depth of the mine, H is known.

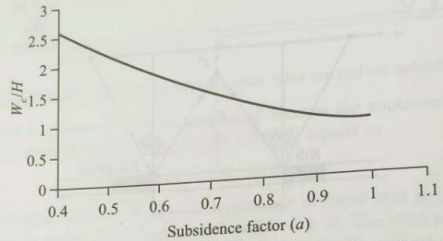


FIGURE 12.7 Relationship between critical width and subsidence factor (adapted from Peng, 1992).

Critical width can also be obtained by identifying "rock mass factor" as defined in Table 12.2 and using Eq. 12.9 (Saxena, 2003).

$$W_c = (2 - M)H \tag{12.9}$$

TABLE 12.2 Rock Mass Factor Based on Condition Roof Strata

Sl. no.	*Nature of strata*	*Rock mass factor* (M)
1.	Totally disturbed roof strata (as in case of thick seam mining)	1.0
2.	Partially disturbed roof strata with thickness of intact strata being less than five times the seam thickness under extraction	0.9
3.	Partially disturbed roof strata with thickness of intact strata being more than five times the seam thickness under extraction	0.8
4.	Naturally weak and disturbed roof strata (having clay, shale or mudstone)	0.7
5.	Roof strata having interbedded layers of sandstones and shales	0.6
6.	Roof strata having predominantly sandstone beds	0.5

EXAMPLE 12.1 Given a longwall panel with following geological and mining data:

Rock type	*Hardness*	*Thickness* (m)
Clay	Regular	12
Shale	Hard	10
Sandstone	Very hard	35
Sandy shale	Hard	15
Shale	Regular	12
Sandstone	Soft	15
Shale	Very Soft	2
Coal	Hard	4

If the mined out length and width of the active panel are 250 m and 100 m, determine the maximum possible subsidence at current time. What will be the maximum subsidence over that panel?

Solution: The following table provides calculations of necessary parameters.

Rock type	*Thickness* (m)	*Hardness*	Q_i	$(h_i Q_i)^2$	$(h_r)^2$
Clay	12	Regular	0.92	121.882	144
Shale	10	Hard	0.40	16.000	100
Sandstone	35	Very hard	0.02	0.490	1225
Sandy shale	15	Hard	0.06	0.810	225
Shale	12	Regular	0.60	51.840	144
Sandstone	15	Soft	0.20	9.000	225
Shale	2	Very Soft	0.80	2.560	4
Coal	4	Hard	0.87		
				202.58	2067

From the above table, we estimate

$$P = \sqrt{\frac{202.58}{2067}} = 0.313,\ a = 0.35 + 0.5 \times 0.313 = 0.506 \text{ and } s_{max} = 0.506 \times 4 = 2.024 \text{ m}$$

Depth of coal seam is 101 m. Using Figure 12.7, we find $W_c = 222$ m.

$$s_o = 2.024\sqrt{\frac{100}{222}} = 1.358 \text{ m}$$

Holla and Barclay, 2000 published a handbook on subsidence and proposed methods for estimating maximum subsidence based on mining induced subsidence data in the New South Wales, Australia. The subsidence data and resulting graphs in this method were obtained from collieries in the area between the Illawarra Escarpment and the Burragorang Valley

(Keilich, et al., 2006). These data were collected over a period of thirty years. The relationship between s_{max}/m and W/H for single panels is given in Figure 12.8. The figure shows that the largest longwall W/H ratio falls into the sub-critical category ($W/H < 1.4$) since deep mining conditions exist in the Southern Coalfield.

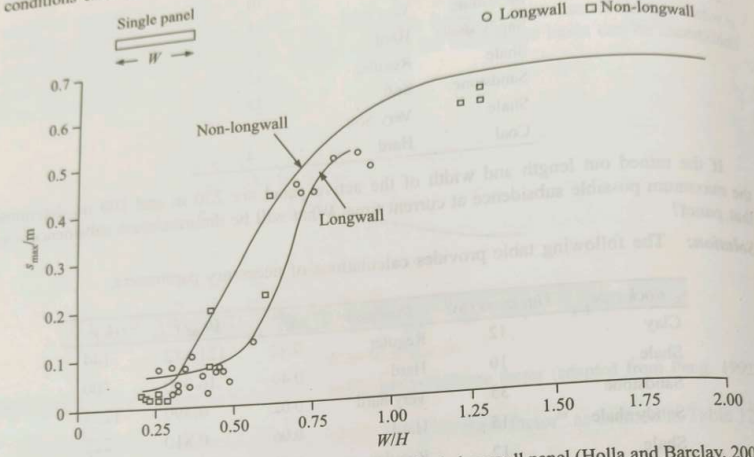


FIGURE 12.8 Relationship between s_{max}/m and W/H for single longwall panel (Holla and Barclay, 2000).

12.4 METHODS OF PREDICTING SUBSIDENCE

There are several techniques to predict mining induced surface subsidence. Each of this technique assumed that the maximum possible surface subsidence cannot exceed the height of extracted seam or mining height. Following are the six basic methods for prediction of mine subsidence:

(i) Empirically derived relationship or graphical method
(ii) Analytical method
(iii) Profile function
(iv) Influence function
(v) Physical models
(vi) Numerical method

12.4.1 Empirical or Graphical Method

These relationships are established from observed surface subsidence behaviour. These relationships require sufficiently large number of observations and case studies in order to establish level

of accuracy for prediction. As for example, the UK SEH (Subsidence Engineers' Handbook) empirical model is based on 200 sites in several coalfield of UK (Whittaker and Reddish, 1989). The subsidence data collected from these sites are rationalised by NCB (National Coal Board) and published in 1965 and 1975. The SEH empirical models are developed based on charts, graphs and tables and relate depth, mining height and width of the panel with maximum surface subsidence and also to generate subsidence profile, strain profiles and others. In general, the maximum possible subsidence, the maximum possible compressional and tensile strain and the maximum slope are obtained as function of depth, mining height and width of the panel as:

$$\frac{s_{max}}{m} = f\left(\frac{W}{H}\right) \tag{12.10a}$$

$$\varepsilon_{max} = g\left(\frac{s_{max}}{H}\right) \tag{12.10b}$$

$$\psi_{max} = h\left(\frac{s_{max}}{H}\right) \tag{12.10c}$$

The detail description of this method is beyond the scope of this chapter. Interested readers can refer to excellent references as Peng, 1992 and Whittaker and Reddish (1989).

12.4.2 Analytical Method

In this method, subsidence is considered as a problem which obeys laws of elasticity, plasticity and visco-elasticity. It is a mathematical expression that gives the value of the field variable at any location in the body. Their use is restricted to situations characterised by relatively simple geometry, boundary conditions and material behaviour.

For problems involving complex shapes, material properties and complicated boundary conditions, it is difficult and in many cases intractable (not easily controlled) to obtain analytical solution that satisfies the governing differential equations or gives extreme value to the governing functions. In order to overcome this limitation, numerical methods are nowadays frequently adopted for analyses related to complex design problems as mentioned below.

It is rarely found that analytical solution can be found for rock mechanics problems of practical concern. This may be because of any one reason or their combinations

(i) The boundary conditions cannot be described by simple mathematical functions
(ii) The governing partial differential equations are non-linear
(iii) The problem domain is inhomogeneous
(iv) The constitutive relations for the rock mass are non-linear
(v) Insufficiently simple mathematical calculation

12.4.3 Profile Function

The profile function $S = f(x)$, x being the distance either from infection point or from the centre of the panel, approaches to predict mining subsidence. This method is probably the

most widely applied of all methods in view of its representation being conveniently expressed in mathematical terms or data-base form. This method predicts profile in set directions across an excavation. It uses the geometry of the excavation along with equations or tables of data to predict a longitudinal or transverse profile for subsidence. This empirical approach is extensively used because it is relatively easy to use and apply to new situations. Profile functions are divided into two types:

(i) Use of equations to define the profile in terms of mining geometry
(ii) Use of table of data to define the profile in terms of mining geometry

The constants for the equations are all derived empirically from local observations. The methods vary in complexity depending upon how many of the more significant factors are taken into account.

(i) Simple method can consider only depth
(ii) Advanced method can consider many parameters

A profile function simply defines the shape of the curve, and it requires a maximum value of subsidence (s_{max}) and a number of constants to calculate subsidence profile. The method is generally applied by adjusting the constants in the equation until a satisfactory shape is obtained which provides an acceptable match to the observed data (Whittaker and Reddish, 1989). Consequently, using this calibrated equation, it should then be possible to predict subsidence curve for that locality. As the profile functions are expressed in the form of equation, values of subsidence can be quickly and easily calculated by computer program. When very little measured data are available, the method can still applied using limited data to calculate the constants. The predicted subsidence profile by this method may not match exactly the shape given by measured data consequently introduce an error. The shape of profile may vary depending upon various external parameters not taken into account by the function. Following are some of the most commonly used profile functions:

Functions based on transition point as origin as shown in Figure 12.9(a):

(i) Hyperbolic function (UK: King and Whetton, 1957, Wardel, 1965, Cherny, 1966, Peng and Chen, 1981)

For critical:
$$S = \frac{s_{max}}{2}\left(1 - \tanh\left(\frac{cx}{R}\right)\right) \tag{12.11}$$

For subcritical:
$$S = \frac{s_{max}}{2}\left(\tanh\left(\frac{n(x+W)}{R}\right) - \tanh\left[\frac{nx}{R}\right]\right) \tag{12.12}$$

where, c and n are arbitrary constants depending on the mine site. Generally, value of n is 2.

(ii) Hyperbolic function (Poland: Knothe, 1953)

$$S = \frac{s_{max}}{2}\left(1 - \left[\frac{2}{\sqrt{\pi}}\int_0^{\pi x R} \exp(-u^2)\,du\right]\right) \tag{12.13}$$

(iii) Exponential function (Hungary: Martos, 1958, Marr, 1958–59)

$$S = s_{max} \exp\left(-\frac{1}{2}\left[\frac{x+R}{R}\right]^2\right) \quad (12.14)$$

(iv) Exponential function (USA: Peng and Chen, 1981)

$$S = s_{max} \exp\left(-\left[\frac{cx}{R}\right]^d\right) \quad (12.15)$$

where, c and d are arbitrary constants depending on the mine site.

(v) Donets trigonometric function (USSR: Geological Institute of Mine Surveying, Anon, 1958)

For critical: $$S = \frac{s_{max}}{2}\left(1 - \frac{x}{R} - \frac{1}{\pi}\sin\left(\frac{\pi x}{R}\right)\right) \quad (12.16)$$

For subcritical: $$S = a_{max}(n_1, n_2)^{1/2}\left(n^2\left[1 - x + \frac{\sin 2\pi x}{2\pi}\right] + \frac{1-n^2}{4}[1+\cos\pi x]^2\right) \quad (12.17)$$

where, n_1, n_2 = coefficient related to width/depth ratio of either side of an inclined seam $n = n_1$ or n_2 depending on side of the panel.

(vi) Trigonometrical function (Hoffman, 1964)

$$S = s_{max} \sin^2\left(\frac{\pi}{4}\left[\frac{x}{R} - 1\right]\right) \quad (12.18)$$

Functions based on face centre origin as shown in Figure 12.9(b)

(i) Polish profile function (Kowalczyk, 1968)

$$S = s_{max} \exp(-\bar{n}x^2) \quad (12.19)$$

where $\bar{n} = \frac{s_{max}}{R^2 C}$; C = Average roof bed lowering.

(ii) Hungarian profile function (Martos, 1958)

$$S = s_{max} \exp\left(-\frac{x^2}{2d^2}\right) \quad (12.20)$$

d = distance of transition point from panel centre

(iii) Niederhofer's method (Niederhofer, 1962)

$$S = s_{max}\left(1 - \left[\frac{x}{P}\right]^2\right)^2 \quad (12.21)$$

(iv) Indian profile function (Kumar, Saxena and Singh, 1983)

For subcritical:
$$S = s_{max} \exp\left(-\left[\frac{nx^2}{P^2 - x^2}\right]\right) \tag{12.22}$$

For critical:
$$S = s_{max} \exp\left(-\left[\frac{nx^4}{P^4 - x^4}\right]\right) \tag{12.23}$$

where n = constant (1 to 5 generally), x = distance from extraction centre, R = radius of critical area of extraction and P = half width of subsidence profile, i.e., $W/2 + R$

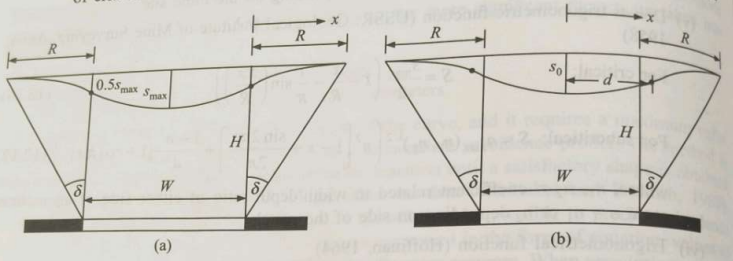
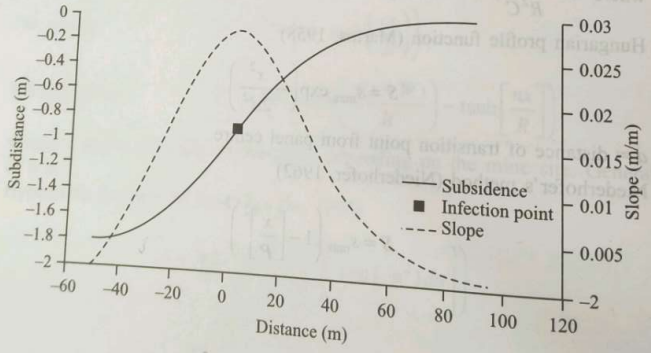


FIGURE 12.9 Parameters for profile function: (a) Critical subsidence, and (b) Subcritical subsidence.

EXAMPLE 12.2 Using data of Example 12.1, determine subsidence and slope profiles on the surface for angle of draw of 35°. Use hyperbolic function with $n = 2$. Also determine the location of inflection point.

Solution: Based on the data given in Example 12.1 the following plot is generated to describe subsidence, slope and inflection point. The data are obtained using Eq. 12.12. It can be noted that the maximum value of subcritical subsidence is found to be 1.798 m as opposed to 1.358 m as found by approximating using Eq. 12.8.



Assuming location of infection point is d distance apart from the origin, we find

$$S(-W/2) = 2 \times S(d)$$

$$\Rightarrow \quad \tanh\left(\frac{2(d+W)}{R}\right) - \tanh\left[\frac{2d}{R}\right] = \tanh\left[\frac{W}{R}\right]$$

Solving the above equation iteratively, we find $d = 3.763$ m.

12.4.4 Influence Function Method

Influence function method utilises the basic theory of superposition to determine the overall influence of an infinitesimal extraction area. Excavation opening is composed of finite number of small openings or elements. Each small opening produces an individual surface subsidence profile given by an influence function and its summation is the total subsidence. Since the overall subsidence is the summation of influence of each mined out elements, it is not necessary to have contiguous excavation like longwall panel. In other words, this method is well suited for determining surface subsidence caused due to bord and pillar panel.

It is based on the theory that an area of influence develops around a point on the surface due to extraction of a small element within the sphere of influence as shown in Figure 12.10. The influence is maximum when element is directly below the surface point being considered. It becomes the minimum when the element is near the edge.

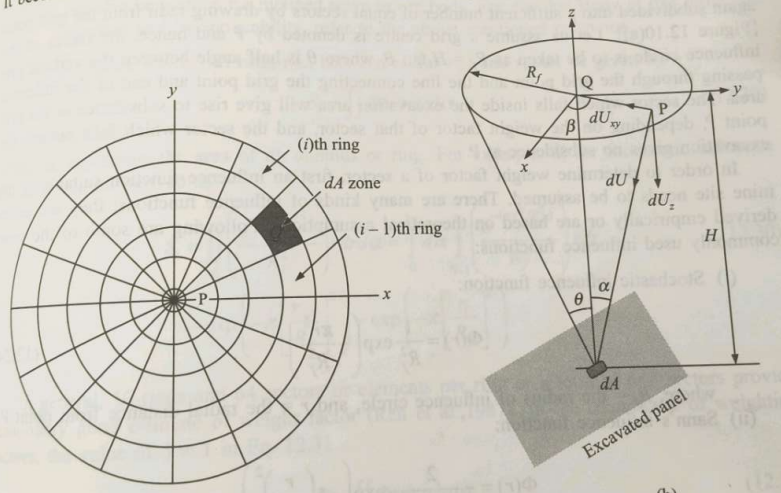


FIGURE 12.10 (a) Sectors of grid point P showing excavated zone dA, and (b) Displacement vector at point P due to excavation of element dA.

A typical influence function would look at annular zones around a point and use equations to relate the percentage extraction in each area to the subsidence of that point. The area of influence is generally assumed to be circular in shape, although hexagonal shapes have been employed occasionally. The functions commonly used can be subdivided as follows:

(i) Empirically derived functions which use arbitrary constants to produce the desired value and form of subsidence

(ii) Functions relating to the rock properties and mechanics of the problem rather than the arbitrary constants.

Second of these groupings is more popular but it is less flexible when carrying out calibration work.

The advantage of this method is the ability to predict subsidence movement at any point above extraction of any practical shape. Drawback of these method is that it is more difficult to apply than profile functions and much more difficult to check and calibrate.

Influence Functions and Weight Factors

The detailed procedure of influence function method is aptly available in various literatures (Peng, 1992; Singh, 1984; Whittaker and Reddish, 1989). The method consists of laying a suitable square or rectangular grid on the surface covering (i) the extraction panel area, and (ii) extending up to the subsidence draw limits. An influence area of circular shape at each grid point (middle of the grid) is divided into a sufficient number of concentric rings. Each ring is again subdivided into a sufficient number of equal sectors by drawing radii from the grid point [Figure 12.10(a)]. Let us assume a grid centre is denoted by P and hence, the radius of the influence circle is to be taken as $R_f = H \tan \theta$, where θ is half angle between the vertical line passing through the grid point and the line connecting the grid point and end of the influence area. The sector which falls inside the excavation area will give rise to subsidence at the grid point P, depending on the weight factor of that sector, and the sector which falls outside the excavation gives no subsidence at P.

In order to determine weight factor of a sector, first an influence function suitable to the mine site needs to be assumed. There are many kinds of influence functions; they are either derived empirically or are based on theoretical assumptions. Following are some of the most commonly used influence functions:

(i) Stochastic influence function:

$$\Phi(r) = \frac{1}{R_f^2} \exp\left(-\frac{\pi r^2}{R_f^2}\right) \tag{12.24}$$

where, R_f = the radius of influence circle, and r = the radial distance from point P.

(ii) Sann's influence function:

$$\Phi(r) = \frac{2}{(\pi)^{3/2} R_f r} \exp\left(-4\left(\frac{r}{R_f}\right)^2\right) \tag{12.25}$$

(iii) Ehrhardt and Sauer influence function:

$$\Phi(r) = \frac{4.6}{\pi R_f^2} \exp\left(-4.6\left[\frac{r}{R_f}\right]^2\right) \tag{12.26}$$

(iv) USBM or Bal's influence function (1932/33):

$$\Phi(\zeta) = \frac{i}{H^2} \cos^2 \zeta \tag{12.27}$$

where, i = unit extraction, ζ = zone angle, which is in the range $0 \le \zeta \le \delta$ (angle of draw). For detailed description of this influence function, refer Peng (1992).

(v) n-k-g influence function for roof having strong floor (Nicieza et al., 2005)

$$\Phi(r) = \frac{Q}{(1+4n)(kr)^2}\left\{\exp\left(-\pi\left[\frac{H}{rk}\right]^2\right) + n\exp\left(-\pi\left[\frac{H}{rk}\right]^2 \Big/ 4\right)\right\} \tag{12.28}$$

where Q is a factor related to the volume of the excavated element and H is the depth of the excavated element, n and k are the two parameters that characterise the ground and that allow the shape and magnitude of the subsidence trough to be adjusted (Nicieza et al., 2005). The parameter n acts mainly on the magnitude of the subsidence, while k allows the shape of the trough to be fitted. The influence function can also be applied for inclined seam or ore body. For detailed study of this influence function and its applicability, refer to the above mentioned reference.

The weight factor for the ith ring, i.e., between radius r_{i-1} and r_i can be given by

$$S_i = \iint_{A_i} \Phi(r) r dr d\alpha \tag{12.29}$$

where A_i represents the area of ith annulus or ring. For example, S_i for Stochastic influence function can be derived as

$$S_i = \iint_{A_i}\left(\frac{e^{-(\pi r^2/R_f^2)}}{R_f^2}\right) r dr d\alpha = \int_0^{2\pi} d\alpha \int_{r_{i-1}}^{r_i}\left(\frac{e^{-(\pi r^2/R_f^2)}}{R_f^2}\right) r dr \tag{12.30}$$

$$= \exp\left(-\pi\left[\frac{r_{i-1}}{R_f}\right]^2\right) - \exp\left(-\pi\left[\frac{r_i}{R_f}\right]^2\right)$$

In general, 10 rings and 64 sectors or elements per ring or a total of 640 sectors provide reasonably good estimate of weight factor (Ren et al.,1987). In the calculation of weighting factors, the value of $S \neq 1$ in Eq. 12.31.

$$S = \iint_{\Omega} \Phi(r)\, r dr d\alpha \tag{12.31}$$

where, Ω = the whole area of the influence circle. However, the value of S must be unity and hence, it requires adjustment of weight factor of each ring as $S_i \geq S_i/S$. It may be noted that at ith ring, each sector has equal adjusted weight factor equals to $S_{in} = S_i/N_i$ where N_i denotes the number of sectors in i^{th} ring. Table 12.3 lists the weight factor of a sector in each ring for stochastic function. Note that if excavation is made in any sector, weight factor is added to grid P. The subsidence at point P is simply the sum of weight factors of all the element which fall in the extraction panel multiplied by the maximum possible subsidence, s_{max}. The weighting factor for the elements falling outside the extracted panel is zero. Hence, the total subsidence at grid point P will be

$$U_z = s_{\max} \sum_{i=1}^{I} \sum_{n=1}^{N_i} S_{in} \tag{12.32}$$

where I signifies number of rings. Note that S_{in} is non-zero quantity even if a sector is extracted partially. For partial extraction, S_{in} can be multiplied by (A/A_s), where A is the extraction area inside the sector and A_s is the area of the sector.

TABLE 12.3 Value of S_i for ith Ring for Stochastic Function

Ring, i	1	2	3	4	5	6	7	8	9	10
S_i	0.03232	0.09109	0.13398	0.15551	0.15571	0.13923	0.11309	0.08424	0.05791	0.03687

Horizontal Displacement and Strain

Let us consider Figure 12.10 again. Assume that a vertical subsidence has occurred at grid point P due to extraction of an element dA in underground. The centroid of dA projected on the surface marks the point Q. Let us assume that the element dA lies inside any of the sectors of the influenced circle of grid point P. Then the displacement vector dU at grid point P has two components viz. (i) vertical component dU_z obtained by the weight factor, and (ii) horizontal component dU_{xy} in $x - y$ plane measured as given by Eq. 12.33.

$$dU_{xy} = dU_z \tan \alpha \tag{12.33}$$

where α is the angle between the vertical and the line joining the element centroid to the grid point P on the surface. It can be measured as:

$$\tan \alpha = \text{PQ}/H \tag{12.34}$$

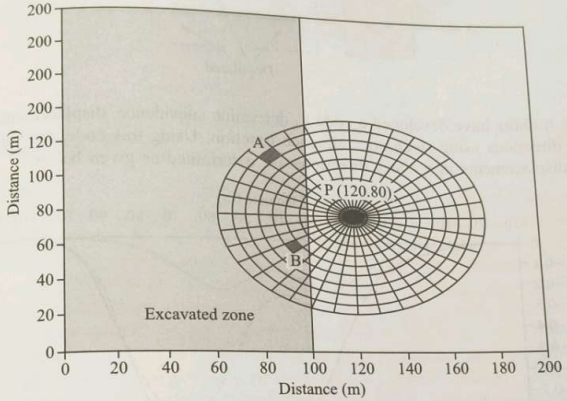
It may be noted that if the circular area fully encompasses the extraction panel, then horizontal displacement at grid point P will be zero. The total horizontal movement at point P is again the sum of all extracted elements as $U_{xy} = \sum_N dU_{xy}$, where N = number of sectors in the influenced zone of grid point P. The resultant dU_{xy} can be resolved in x and y directions as

$$dU_x = dU_{xy} \cos \beta \tag{12.35a}$$

$$dU_y = dU_{xy} \sin \beta \tag{12.35b}$$

where β is the angle between dU_{xy} and x-axis shown in Figure 12.10(b) and can be estimated as $\cos\beta = (x_Q - x_P)/PQ$. The total displacement at P is again a sum of all extracted elements. The slope and strain at point P are then readily calculated having known vertical and horizontal movements.

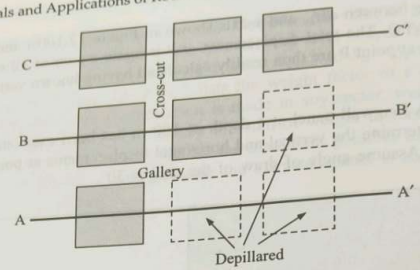
EXAMPLE 12.3 A longwall panel at a depth of 100 m has been excavated as shown in the following figure. Determine the vertical and horizontal displacements at point P for excavation of sectors A and B. Assume angle of draw of the area is 30°.



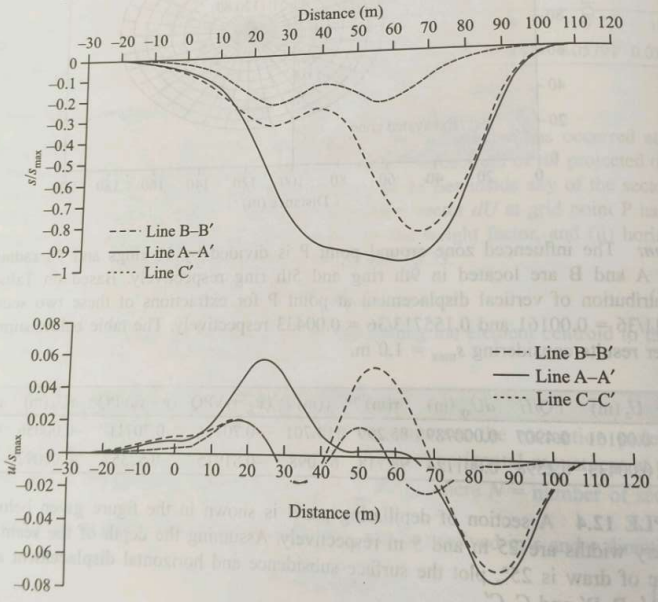
Solution: The influenced zone around point P is divided by 10 rings and 36 radials. The sectors A and B are located in 9th ring and 5th ring respectively. Based on Table 12.3, the contribution of vertical displacement at point P for extractions of these two sectors are 0.057911/36 = 0.00161 and 0.155713/36 = 0.00433 respectively. The table below summarises the other results considering s_{max} = 1.0 m.

Sector	U_z (m)	PQ/H	dU_{xy} (m)	x(m)	y(m)	$(x-x_P)$/PQ	$(y-y_P)$/PQ	dU_x (m)	dU_y (m)
A	0.00161	0.4907	0.000789	85.299	114.701	–0.70711	0.70711	–0.00056	0.00056
B	0.00433	0.2598	0.001124	98.718	65.098	–0.81915	–0.57357	–0.00092	-0.00065

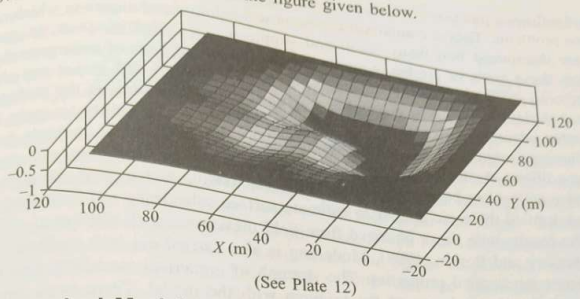
EXAMPLE 12.4 A section of depillaring panel is shown in the figure given below. Pillar and gallery widths are 25 m and 5 m respectively. Assuming the depth of the seam is 50 m and angle of draw is 25°, plot the surface subsidence and horizontal displacement along the lines A–A′, B–B′ and C–C′.



Solution: Authors have developed a code to determine subsidence, displacements and strains in x and y directions using stochastic influence function. Using this code, the subsidence and horizontal displacements in x and y directions are determined as given below.



The overall subsidence is shown in the figure given below.



(See Plate 12)

12.4.5 Physical Model

Physical models have played an important role in the development of knowledge pertaining to understanding ground behaviour in relation to mining subsidence. King and Whetton (1957) reported the finding of mining subsidence investigations using physical models constructed of solidified gelatin solution. Sand/plaster physical models are the most commonly used since other types do not readily provide the advantages given by such modelling materials. In particular, sand/plaster types easily allow the principle variations in geology to be built into the model. Additionally the propagation and growth of cracks caused by caving and subsidence can be observed and studied in relation to other factors. Surface subsidence can be studied as a separate feature, although it is usual to take account of sub-surface behaviour with such models. In this case, the rock strength can be readily adjusted by selecting appropriate sand to plaster ratio in the mixing stage of model construction.

The use of physical models, both two- and three-dimensional were popular from 1950s to the early 1980s using design of equivalent materials and equivalent load. Construction of the models have sometimes involved elaborate simulation of the sedimentary deposition process and the process is somewhat cumbersome. The use of physical models has declined greatly in recent years, due to:

(i) Their high cost and labour intensive nature, especially where several variations of the rock properties and loading conditions are required
(ii) The increasing capability of computer based numerical models to include the above-mentioned features, and to show these effects visually by computer graphics
(iii) The materials usually designed with scaled elastic properties may not realistically model and non-linear inelastic behaviour of actual rock mass

12.4.6 Numerical Method

Numerical methods especially finite element method (FEM), discrete element method (DEM) boundary element method (BEM) and others are alternative procedures to analytical method

profile and influence function methods for analysing structures made in rock mass and associated subsidence problems. They are approximate numerical solution techniques in which continuous systems are discretised into many small and simple pieces called *elements*. In geotechnical engineering, these methods are being applied in design and analysis of underground structures such as tunnels, caverns, powerhouse, mine pillars and galleries and stopes and also utilises non-linear material behaviour to determine the subsidence and strains on the surface.

The major advantages of numerical methods reside on its versatility to apply in various problems having irregular shapes and sizes, multiple material models, multiple loading and boundary conditions. As for example, stresses and deformations around any size and shape of underground opening can be analysed under different loading and support conditions. Several authors have applied these techniques to estimate surface subsidence and horizontal strain and compared the results with those obtained from field measurement (Keilich et al., 2006; Guoet al., 2004, Heasley and Barton, 1998). Modelling is also carried out with multiple rock layers having different mechanical properties. The strength of numerical modelling also lies on the resemblance of the physical size of the problem with the model. There is no need to scale down or up the actual structure being analysed in order to obtain the stress and deformation behaviour. The actual engineering properties of rock or other material are used in finite element analysis. On the contrary, for physical modelling of structures, equivalent material properties are assumed with scale down geometries.

12.5 SINKHOLE SUBSIDENCE

Sinkhole type subsidence occurs due to shallow depth abandoned mine workings, natural cavities, excessive withdrawal of ground water and others. Most of the old mines are worked with room and pillar method of mining and possesses great potential of surface subsidence by roof fall/pillar failure. In UK, more than 70,000 old mine workings are reported and some of them may be three centuries old (Whittaker et al., 1989). In the USA, 354 subsidence incidents were reported over Pittsburgh Coal bed, most of them were in the form of sinkholes (Gray et al., 1977). In 1985, Marino et al., reported that both trough and sinkhole type subsidence occurred in Illinois (Whittaker et al., 1989) over shallow depth room and pillar mines. After extensive study of surface subsidence in the USA, Gray et al., commented that the most prevalent subsidence features over abandoned mined land are sinkholes, with depth of sinkhole more than 3 ft, and trough or sags less than 3 ft (Peng, 1992). After studying subsidence incidents in Germany, Kratzsch commented that sudden cave-ins and irregular depressions in the form of sinkholes over near-surface abandoned mines possess a serious risk to the populated area nearby (Kratzsch, 1983). His study also suggests that size of the affected area must be established based on statistical investigations, taking into account type of workings, strength and thickness of roof, fluctuation of ground water and others. In Korea, more than 300 coal mines have been ceased their operations with the economic structural adjustments in early 1990s. With lots of abandoned coal mines, the surface subsidence problem has been coming up to the public, known as one of geohazards (Choi et al., 2004). Their study suggests that several parameters should be emphasised for evaluating the surface subsidence in coal mine area, and these parameters should be dealt with adding an extra weight for realistic analysis. In addition, the sinkhole type

subsidence in urban area, whatever it caused by collapse of galleries in coal mines or in metal mines, can be a death-blow to many structures as well as public welfare.

All of the above mentioned literatures suggest that sinkhole development is a process of collapsed junction having weak/fractured roof and then progression of collapse chimney up to the surface at shallow depth of cover. In general, sinkhole is in the form of conical depression/cavity suddenly appears on the surface. Figure 12.11(a) shows plan of a 4-way junction with a cylindrical sinkhole of diameter D. Figures 12.11(b), (c) and (d) depict progressive failure of roof above the chimney or sinkhole. The major factors which contribute to sinkhole formation are width of mine opening/gallery (w), height of opening (m), depth of cover (H), rock type and thickness of roof, water condition, pillar strength, time elapsed after mining operation has been ceased, inclination of the seam, bulking factor (K) of roof rocks and others. Whittaker et al., (1989) have reported that width of gallery, depth of cover and roof conditions are the primary factors for the determination of sinkhole occurrences and higher chances are associated with lower H, higher W and weak roof conditions. Peng (1992) had also commented on the similar factors which contributed in sinkhole development in the Pittsburgh Coal bed.

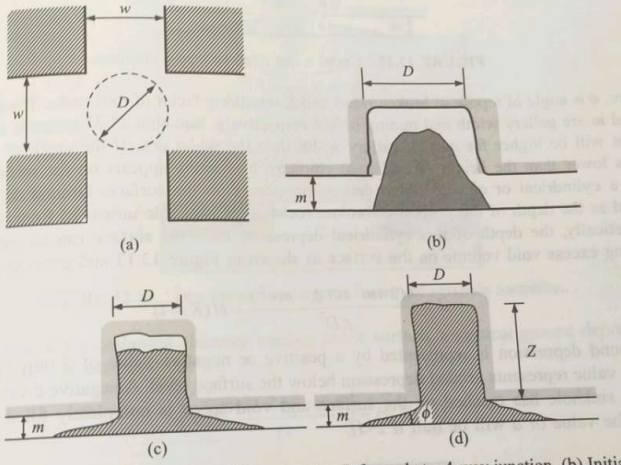


FIGURE 12.11 (a) A circular shape sinkhole of radius D formed at a 4-way junction, (b) Initial collapse of mine roof, (c) Progressive development of roof collapse, and (d) Caved rocks fills completely the void space (adapted from Whittaker and Reddish, 1989).

12.5.1 Volumetric Calculations

In this study, sinkhole is assumed to be cylindrical shape all through its length. Considering the diameter of the chimney or sinkhole as D, the height of collapsed chimney is estimated

by equating the total void space and total volume of broken rock as (Figure 12.12) (Whittaker et al., 1989).

$$V_{caved} = \frac{Kz\pi D^2}{4} \tag{12.36}$$

$$V_{space} = \frac{4(1/2wm^2 \cot\phi) + mw^2 + z\pi D^2}{4}$$

$$\therefore \quad z = \frac{4}{(K-1)\pi D^2}[2wm^2 \cot\phi + mw^2] \tag{12.37}$$

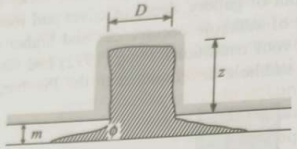


FIGURE 12.12 Caved rocks filled collapsed chimney.

where, ϕ is angle of repose of broken rocks and K = bulking factor of roof rocks. The parameters w and m are gallery width and mining height respectively. Equation 12.37 suggests that caving height will be higher for narrow gallery width than the wider one. If the depth of coal seam (H) is lower than the height of collapsed chimney, a sinkhole appears on the surface. In this case, a cylindrical or conical shape depression occurs on the surface. Ground depression is termed as the depth of top portion of broken rocks in the sinkhole measured from the surface. Theoretically, the depth of this cylindrical depression from the surface can be estimated by equating excess void volume on the surface as shown in Figure 12.13 and given in Eq. 12.38:

$$d = \frac{4(2wm^2 \cot\phi + mw^2)}{\pi D^2} - H(K-1) \tag{12.38}$$

Ground depression is represented by a positive or negative value of d only if $z>=H$. A positive value represents ground depression below the surface level. A negative d value signifies that the sinkhole has reached on the surface and void space is completely filled by broken rocks. The value of d will be null if $z<H$.

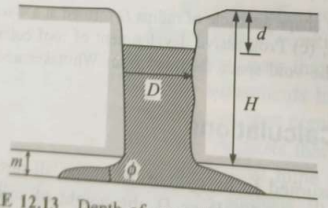


FIGURE 12.13 Depth of ground depression for $z > H$.

From earlier analysis, it is found that the volume occupied by the caved rocks in the adjoining rooms of a 4-way junction ($2wm^2 \cot \phi$) is significant as compared to volume of the junction itself (wm^2) and ratio of these two volumes is:

$$r = \frac{2m \cot \phi}{w} \tag{12.39}$$

Assuming similar area ratio in two-dimensional analyses, the total void area can be estimated as (Figure 12.14)

$$A_{space} = mw[1 + r] + A_p = mw\left[1 + \frac{2m \cot \phi}{w}\right] + zD \tag{12.40}$$

where A_p is the area released by the broken rocks in the roof. Hence, equating area of void space and area occupied by the broken rocks, the height of cavity will be

$$z = \frac{mw + 2m^2 \cot \phi}{D(K-1)} \tag{12.41}$$

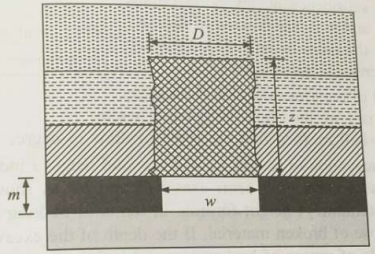


FIGURE 12.14 Two dimensional approximation of sinkhole formation.

As before, if the collapsed chimney reaches to the surface, a vertical ground depression is developed and it is estimated as:

$$d = \left[\frac{mw + 2m^2 \cot \phi}{D}\right] - H(K-1) \tag{12.42}$$

EXAMPLE 12.5 A chimney cave forms over a square excavation of side 6 m and height 3 m. The caved zone forms an inscribed circular cross-section. If the bulking factor is 1.2 and angle of repose of broken material is 40°, how high will cave zone extend above the excavation?

Solution: Based on Eq. 12.37, we estimate the height as:

$$z = \frac{4}{(K-1)\pi D^2}[2wm^2 \cot \phi + mw^2] = 251.94 \text{ m}$$

From earlier analysis, it is found that the volume occupied by the caved rocks in the adjoining rooms of a 4-way junction ($2wm^2 \cot \phi$) is significant as compared to volume of the junction itself (wm^2) and ratio of these two volumes is:

$$r = \frac{2m \cot \phi}{w} \tag{12.39}$$

Assuming similar area ratio in two-dimensional analyses, the total void area can be estimated as (Figure 12.14)

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where A_p is the area released by the broken rocks in the roof. Hence, equating area of void space and area occupied by the broken rocks, the height of cavity will be

$$z = \frac{mw + 2m^2 \cot \phi}{D(K-1)} \tag{12.41}$$

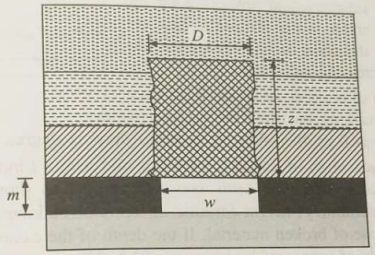


FIGURE 12.14 Two dimensional approximation of sinkhole formation.

As before, if the collapsed chimney reaches to the surface, a vertical ground depression is developed and it is estimated as:

$$d = \left[\frac{mw + 2m^2 \cot \phi}{D}\right] - H(K-1) \tag{12.42}$$

EXAMPLE 12.5 A chimney cave forms over a square excavation of side 6 m and height 3 m. The caved zone forms an inscribed circular cross-section. If the bulking factor is 1.2 and angle of repose of broken material is 40°, how high will cave zone extend above the excavation?

Solution: Based on Eq. 12.37, we estimate the height as:

$$z = \frac{4}{(K-1)\pi D^2}[2wm^2 \cot \phi + mw^2] = 251.94 \text{ m}$$